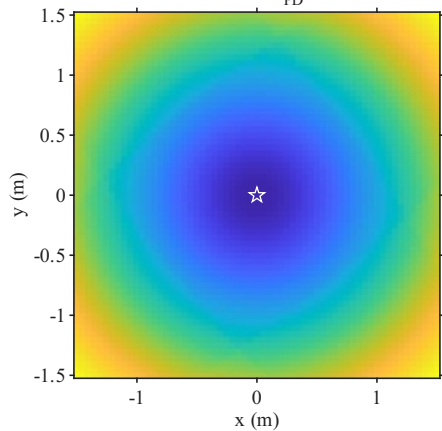


Common color scale; gray: singular or nonregular; star: LED projection

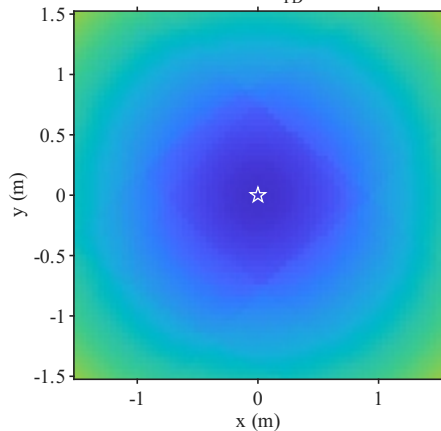
$\Phi_{1/2} = 30$ deg

$z = 0.8$ m; $K = 14$; $A_{PD} = 26.4$ mm²



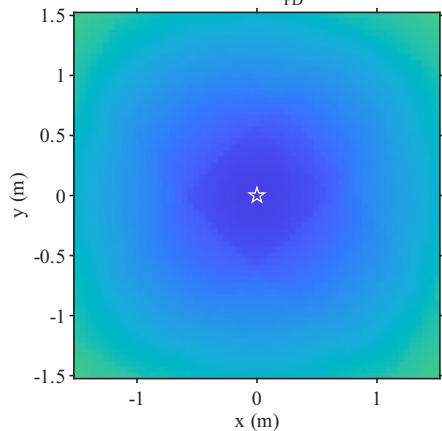
$\Phi_{1/2} = 45$ deg

$z = 0.8$ m; $K = 14$; $A_{PD} = 26.4$ mm²



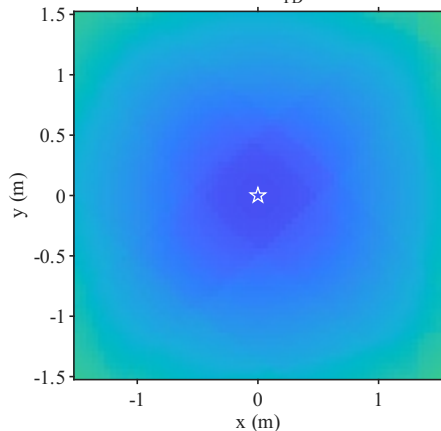
$\Phi_{1/2} = 60$ deg

$z = 0.8$ m; $K = 14$; $A_{PD} = 26.4$ mm²



$\Phi_{1/2} = 75$ deg

$z = 0.8$ m; $K = 14$; $A_{PD} = 26.4$ mm²



3D PEB (cm)

10^1

10^0